The Lagrange kernel ψ is algebraic of degree exactly 5

Day 151: a closed form from the master curve, two failed predictions, and what this does *not* do for Conjecture P

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Grade of the main result: `computed, not proved`. See "Why this is `computed` and not `proved`" at the end of §1 — the derivation in §2 takes two Day-149 statements as *given* rather than proving them.

1 Result

The object

Let u_1, u_2, u_3 be indeterminates, $E_i = e_i(u)$, $V = \prod_{i<j}(u_i - u_j)$, $\mathcal{T}^+ : u^\alpha \mapsto \prod_i u_i^{(\alpha_i)}$ (rising factorials), $\Psi^+(f) = \mathcal{T}^+(fV)/V$, and

$$F_P = \sum_{b\geq 0} P_b \frac{T^b}{b!} = \frac{\mathcal{T}^+(e^{Te_2}V)}{V}, \qquad H = \frac{\tau(F_P)}{F_P}, \qquad \tau : u_i \mapsto u_i + 1.$$

With $\mathrm{wt}(E_1^a E_2^b E_3^c) = a+2b+3c$, let $\mathcal{W}_n$ be the wt $= n$ part of $[T^n]H$ (this is the top homogeneous u-component, by (H2)), $W_n := \mathcal{W}_n/(n+1)$, and

$$\ell_0^{\mathrm{top}}(H) = \sum_{n\geq 0} \mathcal{W}_n T^n = \sum_{n\geq 0}(n+1) W_n T^n = \frac{dY}{dT}, \qquad Y = T\,\mathcal{W}(T), \qquad \mathcal{W}(T) = \sum_n W_n T^n.$$

Then ψ is the Lagrange kernel of Y: writing $T(\cdot)$ for the compositional inverse,

$$\psi(Y) := \mathcal{W}(T(Y)) = Y/T(Y), \qquad \text{equivalently} \qquad Y = T\,\psi(Y).$$

This is layer $d = 0$ of Conjecture P: at $E_3 = 0$ the W_n are the Narayana polynomials, and E-positivity of ψ is the top layer of the Day-150 layer-induction attack.

The result

Theorem 1 (`computed`). *ψ is algebraic over $\mathbb{Q}(E_1, E_2, E_3)(Y)$ of degree **exactly** 5. Explicitly, let ν_i solve $\nu_i(1 - T(e_1(\nu) - \nu_i)) = u_i$, let $P := e_1(\nu)$, $q := 1 - TP$, and $\Delta_2 := E_1^2 - 4E_2$. Then*

$$\boxed{\psi \;=\; q\,\frac{e_3(\nu)}{E_3} \;=\; q\prod_{i=1}^{3}\frac{\nu_i}{u_i} \;=\; \frac{q}{\prod_{i=1}^{3}(q + T\nu_i)}}$$

and, purely in the curve coordinate q,

$$\boxed{\psi \;=\; \frac{q(1 - 2E_1T + \Delta_2T^2 - q^2)}{4E_3T^3(q+2)}, \qquad Y = T\psi}$$

where q is the branch at $q = 1 + O(T)$ of the master curve $\sum_i \sqrt{q^2 + 4Tu_i} = q + 2$.

Eliminating q gives the minimal polynomial. Write $Q(\psi, Y, E) = \sum_{j=0}^{9} Y^j C_j(\psi, E)$ with

$$\begin{aligned}
C_0 &= \psi^4(\psi - 1)\\
C_1 &= -E_1\psi^3(5\psi - 4)\\
C_2 &= \psi^2\big(10E_1^2\psi - 6E_1^2 - E_2\psi^2 - 8E_2\psi + 8E_2\big)\\
C_3 &= -2\psi\big(5E_1^3\psi - 2E_1^3 - 2E_1E_2\psi^2 - 12E_1E_2\psi + 8E_1E_2 - 11E_3\psi^2 + 44E_3\psi - 32E_3\big)\\
C_4 &= 5E_1^4\psi - E_1^4 - 6E_1^2E_2\psi^2 - 24E_1^2E_2\psi + 8E_1^2E_2 + E_1E_3\psi^3 - 36E_1E_3\psi^2 + 80E_1E_3\psi\\
&\quad + 8E_2^2\psi^2 + 16E_2^2\psi - 16E_2^2\\
C_5 &= -E_1^5 + 4E_1^3E_2\psi + 8E_1^3E_2 - 3E_1^2E_3\psi^2 + 6E_1^2E_3\psi + 8E_1^2E_3 - 16E_1E_2^2\psi - 16E_1E_2^2\\
&\quad - 20E_2E_3\psi^2 + 104E_2E_3\psi - 32E_2E_3\\
C_6 &= -E_1^4E_2 + 3E_1^3E_3\psi + 8E_1^3E_3 + 8E_1^2E_2^2 + 4E_1E_2E_3\psi - 32E_1E_2E_3 - 16E_2^3\\
&\quad - E_3^2\psi^2 + 88E_3^2\psi - 16E_3^2\\
C_7 &= -E_3\big(E_1^4 - 16E_1^2E_2 - 20E_1E_3\psi + 16E_1E_3 + 48E_2^2\big)\\
C_8 &= 8E_3^2\big(E_1^2 - 6E_2\big)\\
C_9 &= -16E_3^3.
\end{aligned}$$

Then $Q(\psi, Y, E) = 0$, with

$$\deg_\psi Q = 5, \qquad \deg_Y Q = 9, \qquad Q \textbf{ irreducible} \text{ over } \mathbb{Q}[E_1, E_2, E_3, Y, \psi], \qquad Q \text{ monic in } \psi.$$

Every C_j is E-homogeneous of degree j for $\deg E_k = k$, as forced by the scaling $u \mapsto \lambda u$, $Y \mapsto Y/\lambda$. Irreducibility is `sympy.factor_list`. Q was obtained by *resultant*, not by fitting; a separately fitted bivariate ansatz returns the same equation (§6).

The $E_3 = 0$ degeneration — the sanity check that makes it believable

At $E_3 = 0$ the kernel is known in closed form, $\psi = \prod_i(1 + u_iY) = 1 + E_1Y + E_2Y^2$ (Narayana / Lagrange inversion). And indeed

$$Q\Big|_{E_3=0} \;=\; -(\psi - 1 - E_1Y - E_2Y^2)\big(\psi^2 - 2E_1Y\psi + \Delta_2Y^2\big)^2.$$

The first factor is exactly the known answer. The squared factor is a pair of spurious branches of $\prod_{i<j}(\psi - (u_i + u_j)Y)$ type, introduced by the resultant. The second degeneration is $E_1 = E_2 = 0$, which reproduces the quintic slice equation of §3 on the nose. A degree-5, degree-9 polynomial with 23 nonzero C_j-terms that factors correctly at both degenerations is not an accident.

Why this is `computed` and not `proved`

Say it here, not in a footnote. The derivation in §2 is a complete algebraic computation *given* two statements from Day 149 §4, which I used and did not prove:

1. $\log \ell_0^{\mathrm{top}}(H) = \partial\,\Xi$ with $\Xi := \ell_1^{\mathrm{top}}(\log F_P)$ and $\partial = \sum_i \partial_{u_i}$;
2. $\theta\,\Xi = T\,e_2(\nu) = \frac{1}{2}(P - E_1)$, with $\theta = T\,d/dT$.

Both are verified numerically against F_P built from its definition (§6), and the ℓ^{top} bookkeeping behind (1) is written down nowhere as a referee-proof argument. Until it is, the whole chain is `computed`: the machine agreed on every case tried. Evidence, not proof.

2 Derivation

2.1 The master curve, rationalised

Day 149 Theorem 4: $q = 1 - Te_1(\nu)$ is the branch at $q = 1 + O(T)$ of $\sum_{i=1}^{3} \sqrt{q^2 + 4Tu_i} = q + 2$. Clearing the three radicals by two resultants ($r_3 = q + 2 - r_1 - r_2$, eliminate r_1, then r_2) and symmetrising gives, up to the factor -256,

$$\begin{aligned} P(q,T,E) = {} & (q-1)(q+1)^3(2q+1) - 2E_1T(q+1)^2(q^2-2q-2) \\ & - 2T^2(q+1)\big[\Delta_2(q^2+q+1) + 2E_1^2(q+1)\big] \\ & + 2E_1T^3\Delta_2(q^2+2q+2) + 16E_3T^3(q+2)^2 - T^4\Delta_2^2, \end{aligned}$$

with $\deg_q P = 5$, $\deg_T P = 4$. This reproduces the Day-149 quintic exactly (difference identically zero, symbolically). At $T = 0$ it is $(q-1)(q+1)^3(2q+1)$, so $q = 1$ is a simple branch.

2.2 Two supporting identities

Everything below uses only $\partial u_i = 1$, $R_i := q + 2T\nu_i = \sqrt{q^2 + 4Tu_i}$, $\sum_i R_i = q + 2$, and the single quantity

$$S_1 := \sum_{i=1}^{3} \frac{1}{R_i}.$$

Differentiating $T\nu_i^2 + q\nu_i = u_i$ by ∂ and by d/dT, and using $T\nu_i/R_i = \frac{1}{2} - \frac{q}{2R_i}$ and $\sum_i u_i/R_i = (q + 2 - q^2S_1)/(4T)$:

$$A := \partial P = \frac{2S_1}{qS_1 - 1}, \qquad T\dot{q} = \frac{q^2S_1 - q - 2}{2(qS_1 - 1)}.$$

Proposition 2 (`computed`: derived from (1)–(2) above)**.** $\dfrac{dY}{dT} = \ell_0^{\mathrm{top}}(H) = \dfrac{e_3(\nu)}{E_3} = \displaystyle\prod_{i=1}^{3} \frac{\nu_i}{u_i} = \prod_{i=1}^{3} \frac{1}{q + T\nu_i}.$

Computation. From $\dot{\nu}_i/\nu_i = -(\nu_i + \dot{q})/R_i$ and $n_3 := e_3(\nu)$,

$$T\frac{d}{dT}\log n_3 = -\tfrac{3}{2} + \tfrac{q}{2}S_1 - T\dot{q}\,S_1 = -\tfrac{3}{2} + \tfrac{q}{2}S_1 - \frac{S_1(q^2S_1 - q - 2)}{2(qS_1 - 1)} = \frac{S_1}{qS_1 - 1} - \tfrac{3}{2} = \frac{A - 3}{2},$$

the last step being the one-line identity

$$q(qS_1 - 1) - (q^2S_1 - q - 2) = 2.$$

On the other side, $T\frac{d}{dT}\log \ell_0^{\mathrm{top}}(H) = T\frac{d}{dT}\partial\Xi = \frac{1}{2}(\partial P - 3) = \frac{A-3}{2}$ as well. Both sides are $1 + O(T)$ (at $T = 0$, $\nu_i = u_i$), so they agree. $\square$

Proposition 3 (`computed`). $Y = T\,q\,\dfrac{dY}{dT}$, *hence* $\psi = Y/T = q\,e_3(\nu)/E_3$.

Computation. Set $Z := TqY'$. Then $\dot Z/Y' = q + T\dot q + TqY''/Y'$, and substituting $TY''/Y' = \frac{A-3}{2}$ (Prop. 2) together with $T\dot q$:

$$q + \frac{q^2S_1 - q - 2}{2(qS_1-1)} + q\Big(\frac{S_1}{qS_1-1} - \frac{3}{2}\Big) = -\frac{q}{2} + \frac{(q+2)(qS_1-1)}{2(qS_1-1)} = 1.$$

So $\dot Z = Y'$ and $Z(0) = 0 = Y(0)$, giving $Z = Y$. □

Both propositions collapse to a single one-line cancellation each. That is the whole content: the S_1-dependence cancels, and what is left is q.

2.3 Everything in q alone

$e_2(\nu) = \dfrac{1-q-E_1T}{2T^2}$, and $\prod_i(q+T\nu_i) = q^2 + qT^2e_2(\nu) + T^3e_3(\nu) = \dfrac{q(q+1-E_1T)}{2} + T^3n_3$. Solving the resulting quadratic for n_3 requires $q^2(q+1-E_1T)^2 + 16E_3T^3$ to be a square *modulo the curve*. It is: on $P = 0$,

$$q^2(q+1-E_1T)^2(q+2)^2 + 16E_3T^3(q+2)^2 = S^2, \qquad S = (q+1)(q^2+q+1) - E_1T(q^2+2q+2) + \Delta_2T^2$$

(`sympy.factor` returns an exact square). Hence

$$\frac{dY}{dT} = \frac{n_3}{E_3} = \frac{S - q(q+1-E_1T)(q+2)}{4E_3T^3(q+2)} = \frac{1 - 2E_1T + \Delta_2T^2 - q^2}{4E_3T^3(q+2)}, \qquad \psi = q\cdot\frac{dY}{dT},$$

which is Theorem 1. The apparent E_3 in the denominator is a genuine 0/0: at $E_3 = 0$ the numerator is $4T^2\big(m(1-E_1T-T^2m) - E_2\big)$ with $m = e_2(\nu)$, which vanishes identically because $m = E_2 + E_1Tm + T^2m^2$ there. Not a blunder.

2.4 The elimination

From $\psi = q\,n_3/E_3$ and $\prod_i(q+T\nu_i) = E_3/n_3$, clearing and substituting $T = Y/\psi$ gives a *cubic* in q:

$$R: \qquad q^3\psi^2 + q^2\psi(\psi - E_1Y - 2) + 2E_3Y^3 = 0.$$

Taking $\mathrm{Res}_q\big(P(q, Y/\psi, E)\cdot\psi^{\deg},\, R\big)$ and factoring yields two degree-5-in-ψ factors; the one that restricts correctly at $E_3 = 0$ is Q. The other factor is unexplained — presumably the conjugate branch. I did not chase it.

3 The $E_1 = E_2 = 0$ slice

Here the u_i are the cube roots of E_3, $\Delta_2 = 0$, and the whole chain depends on T, E_3 only through $\sigma := E_3T^3$. The curve collapses to

$$(q-1)(q+1)^3(2q+1) + 16\sigma(q+2)^2 = 0, \qquad e_3(R) = \sqrt{q^6 + 64\sigma}, \qquad \frac{dY}{dT} = \frac{1-q^2}{4\sigma(q+2)}.$$

Set $f := \psi|_{E_1=E_2=0}$ and $W := E_3Y^3$ (so $W = \sigma\psi^3$). Then

n	0	1	2	3	4	5	6	7
$[W^n]\psi$	1	2	5	**34**	334	3958	52599	755256

continuing 11467146, 181608526, 2972696179, 49967412130, 858396503720, 15017677772264, 266829003005636, ... (40 terms computed).

Theorem 4 (computed)**.** *f satisfies*

$$\boxed{f^5 - f^4 + W(22f^3 - 88f^2 + 64f) + W^2(-f^2 + 88f - 16) - 16W^3 = 0}$$

which is irreducible over $\mathbb{Q}$ and is exactly $Q|_{E_1=E_2=0}$ (up to sign).

The slice equation was found by an ansatz fit ($\deg_f \le 5$, $\deg_W \le 3$; 24 unknowns, 26 equations used) returning a 1-dimensional nullspace whose residual vanishes on *all* 40 computed coefficients — an over-determination margin of 16 free checks. Nothing with $\deg_f \le 4, \deg_W \le 7$ or $\deg_f \le 5, \deg_W \le 2$ fits. Newton-solving the boxed equation from scratch regenerates $1, 2, 5, 34, 334, \ldots$ exactly. It then also drops out of Q by specialisation, so it is derived, not merely fitted.

Discriminant and growth.

$$\mathrm{disc}_f = 1728\, W^4 (W^3 + 636W^2 + 1344W - 64)^3.$$

The dominant singularity is the positive root $W^* = 0.04659172\ldots$ of $W^3 + 636W^2 + 1344W - 64$, so $[W^n]\psi$ grows like 21.46^n — consistent with the observed coefficient ratio climbing through 17.8 at $n = 14$.

OEIS. Not in OEIS. Neither $1, 2, 5, 34, 334, 3958, 52599$ nor the associated $\ell_0^{\mathrm{top}}(H)$ slice $1, 8, 119, 2200, 45500, 1007904$ returns anything, on several prefixes. New sequences.

4 Two failed predictions

Negatives are results. Both of these were pre-registered, both failed, and both are recorded here rather than quietly dropped.

4.1 (a) The Catalan prediction: FAIL

Pre-registered on Day 150 (cycle 2), *in writing, before any computation*:

$$[W^n]\psi|_{E_1=E_2=0} = C_{n+1} \quad \text{(Catalan)}, \qquad \text{i.e.} \qquad [Y^9]\psi = 14E_3^3, \quad [Y^{12}]\psi = 42E_3^4.$$

The motivation was symmetry: Catalan at both extremes of the E_3-filtration, since at $E_3 = 0$ the Narayana row sums are Catalan.

n	0	1	2	3	4	5
$[W^n]\psi\|_{E_1=E_2=0}$ (actual)	1	2	5	**34**	**334**	**3958**
C_{n+1} (predicted)	1	2	5	14	42	132

$$[Y^9]\psi = 2E_2^3E_3 + 22E_1E_2E_3^2 + \mathbf{34}E_3^3 \implies [Y^9]\psi|_{E_1=E_2=0} = 34E_3^3 \ne 14E_3^3,$$
$$[Y^{12}]\psi|_{E_1=E_2=0} = 334E_3^4 \ne 42E_3^4.$$

The prediction agreed on *exactly the three terms* $1, 2, 5$ *that were used to generate it*, and diverged at the first genuinely new data point — not marginally: 34 against 14, then 334 against 42. This is a clean instance of fitting a classical sequence to its own defining data, and it carried zero independent evidence. **That is why one pre-registers.** Had the guess been made after seeing the answer, it would never have been written down; had it been made before and not recorded, the failure would have been invisible.

Consequences, per the pre-registration's own branch: the "ψ is Catalan at both ends of the E_3-filtration" story is dead. Narayana at $E_3 = 0$ is untouched (verified to $n = 33$). Also refuted along the way, on the same slice: not Fuss–Catalan, not large Schröder, not Motzkin, not hypergeometric (successive ratios $2, \frac{5}{2}, \frac{34}{5}, \frac{167}{17}, \frac{1979}{167}$ are not a low-degree rational function of n; 167 and 1979 are prime), and *not* Arc A's Lagrange kernel $\Phi_A = 1+9G+58G^2+322G^3+\cdots$ — a second, sharper confirmation of the Day-150 cycle-2 negative that sharing a curve does not mean sharing a kernel.

4.2 (b) The Kerov character polynomial bridge: FAIL

Day 150 hypothesis: Conjecture P's E-positivity is the shadow of Kerov/Féray positivity. Kerov character polynomials express the normalised characters Σ_k in the free cumulants R_j of the transition measure, with non-negative integer coefficients (Féray's theorem) — a positivity theorem about a Kostka-adjacent expansion in Λ^*, proved by counting a different model rather than by a sign-reversing involution. Exactly Conjecture P's shape. The pre-registered load-bearing test was **T1**: does the 3-variable ring $\Lambda_3 = \mathbb{Z}[E_1, E_2, E_3]$ receive the R_j, or does truncation destroy the structure? Pre-registered rule: *if it destroys it, the bridge is decorative and the question closes.*

The frame map, derived first and verified two ways (11 shapes $\times$ 10 partitions exactly; and P_b re-derived in the λ-frame for $b \leq 4$ over 12 partitions), is

$$\mathfrak{s}_\mu(u)\Big|_{u_i=-(\lambda_i+3-i)} = (-1)^{|\mu|}\, s^*_\mu(\lambda),$$

i.e. Rick's u_i are minus the Okounkov–Olshanski shifted coordinates. In E-coordinates $R_2 = -E_1 - 3$, $R_3 = E_1^2 + 5E_1 - 2E_2 + 10$, and so on; the R_j were independently validated by computing Σ_k from scratch (Murnaghan–Nakayama plus hook lengths) for $k \leq 7$.

Verdicts.

- **(a) PASS.** $R_j|_{\Lambda_3} \neq 0$ for $j = 2, \ldots, 8$, with $\deg_u R_j = j-1$ exactly and leading symbol $(-1)^{j-1} p_{j-1}(u)$. Nothing is killed.
- **(b) PASS.** $\det \partial(R_2, R_3, R_4)/\partial(E_1, E_2, E_3) = -6 \neq 0$, so $(E_1, E_2, E_3) \mapsto (R_2, R_3, R_4)$ is a polynomial automorphism of $\mathbb{A}^3_{\mathbb{Q}}$.
- **(c) FAIL over $\mathbb{Z}$.** $E_2 = \frac{1}{2}(R_2^2 + R_2 - R_3) + 2$ and E_3 picks up a 6. Jacobian $-6 \neq \pm 1$, so $\mathbb{Z}[R_2, R_3, R_4] \subsetneq \Lambda_3$, and Rick's ring is integral.
- **(d) FAIL.** Free generation dies at R_5: in three variables $R_5, \ldots, R_8$ are dependent, e.g. $R_5 = \frac{1}{6}R_2^4 - R_2^3 - R_2^2 R_3 + \frac{13}{6}R_2^2 + 5R_2R_3 + \frac{4}{3}R_2R_4 - 6R_2 + \frac{1}{2}R_3^2 - 11R_3 - 6R_4$. So "the expansion in the R's", unique in Λ^*, is *ill-posed* in Λ_3.
- **(e) FAIL — decisive. Kerov positivity does not survive the truncation.** Rewrite the Σ_k themselves, the objects Féray *proved* positive, in the only available generating set R_2, R_3, R_4 of $\Lambda_3 \otimes \mathbb{Q}$:

k	terms	**negative** coefficients	integral?
$1, 2, 3$	$1, 1, 2$	0	yes
4	10	**5**	no (/6)
5	14	**6**	no
6	22	**10**	no
7	26	**12**	no (/36)

$\Sigma_1, \Sigma_2, \Sigma_3$ survive only because they use nothing above R_4.

T2 as stated ("expand P_b in the R's; are the coefficients non-negative?") is therefore ill-posed. The fair repair — lift to the stable Λ^*, where the expansion is unique and where Féray positivity actually lives — also fails, for both natural lifts, at every $b \le 4$ (negatives: 2/3, 5/9 and 5/11, 11/20 and 6/13, 20/39 and 28/56; none integral). It fails at the smallest case *in closed form*:

$$P_1 = s^*_{(1,1)} = \tfrac{1}{2}(R_2^2 - R_2 - R_3).$$

The pipeline is validated, not broken: it reproduces Biane's $\Sigma_1, \dots, \Sigma_7$ exactly with all coefficients non-negative integers, including $\Sigma_7 = R_8 + 70R_6 + 84R_2R_4 + 56R_3^2 + 14R_2^3 + 469R_4 + 224R_2^2 + 180R_2$. So it detects Kerov positivity where it is there. It is not there for P_b.

The tests are *not* circular: they test positivity of Kerov's Σ_k, an external theorem, never Conjecture P itself.

Bridge closed. By my own pre-registered criterion. "Kerov" does not appear in the FPSAC draft. Import scorecard: roughly nine attempts, zero theorems.

The one keeper. T1(a) is a genuine structural fact about my *own* grading, obtained without importing anything: under $u_i = -(\lambda_i + 3 - i)$ the **Kerov filtration and the u-degree filtration coincide**, with $\deg_u R_j = j - 1$, $\mathrm{lead}(R_j) = (-1)^{j-1} p_{j-1}(u)$, Jacobian -6. So the $E_3 = 0$/Narayana top layer sits at Kerov's top layer too. That is worth a line in the object dictionary. The layer-induction route never depended on Kerov and is untouched.

5 What the closed form says about Conjecture P

The good news

ψ is E-positive far past anything previously checked. Every coefficient of ψ_k in the monomial basis $E_1^a E_2^b E_3^c$, for $k \le 33$, is a **non-negative integer**: **364 monomials, zero negatives, zero non-integers**, exact integer arithmetic throughout, with the two divisibility assertions (by $n+1$ and by $b!$) checked rather than assumed. Layer $d = 0$ of Conjecture P now rests on 364 positive monomials instead of the 7 terms that were on the table on Day 149.

Three structural regularities, all verified to Y^{33}:

- ψ_k is **weight-homogeneous of weight** k for $\mathrm{wt}(E_1^a E_2^b E_3^c) = a+2b+3c$. Each ψ_k therefore lives in a finite-dimensional space; the monomial counts grow slowly $(1,1,1,1,1,1,2,2,3,3,4,4,6,5,7,7,9,8,11,10,13,12,15,14,18,16,20,19,23,21,26,24,29,27)$.
- ψ_k is E_3-divisible for $k \ge 3$.
- **The minimal-E_3 monomials are constant with period 2:**

  $$[E_2^m E_3]\,\psi_{2m+3} = 2 \ (m \le 15), \qquad [E_1 E_2^m E_3]\,\psi_{2m+4} = 1 \ (m \le 14).$$

  These two edge families are the first structure in ψ that is not just "the Narayana end". Any combinatorial model has to make them singletons/pairs.

The bad news, and it is the headline of this section

The closed form does not yield positivity. Q is not manifestly positive: its leading term is $-16E_3^3Y^9$, it has mixed signs throughout the C_j table, and it is degree 5. Degree 5 with a negative leading coefficient is not the shape of a "positive Lagrange kernel". A positive algebraic kernel with a combinatorial model should look like a quadratic or a Fuss-type equation with a sign pattern you can read off; this does not.

So: knowing ψ exactly buys the ability to check positivity to any order cheaply, and it buys the E_3-divisibility statement (transparent now: the whole E_3-dependence enters through the single term $16E_3T^3(q+2)^2$ of the curve, so at $E_3 = 0$, ψ truncates to $\prod_i(1+u_iY)$ and every correction is $O(E_3)$; and in the C_j table, C_j for $j \geq 7$ is divisible by E_3, C_9 by E_3^3). It does *not* buy a certificate. Positivity will need a different one.

This is a partial disappointment and I am not going to dress it up. The hope was that identifying ψ would hand over a tree species and the top layer of Conjecture P would fall out by Lagrange inversion. Degree 5, irreducible, ugly Q, sequence not in OEIS, Catalan refuted — a simple tree species now looks *unlikely.* The layer-induction attack survives, but step 0 is still a conjecture, and it is now a conjecture about an object we can write down exactly and still cannot see the positivity of.

6 Verification

Everything below was checked before anything above was believed.

Claim	How	Verdict
Quintic from radical clearing = Day-149 quintic	resultants + symmetrise, symbolic	exact, difference 0
Chain reproduces the seven published coefficients $\psi = 1 + E_1Y + E_2Y^2 + 2E_3Y^3 + E_1E_3Y^4 + 2E_2E_3Y^5 + (E_1E_2E_3 + 5E_3^2)Y^6$	symbolic-E pipeline from the curve	**all seven**
Same seven, *second independent way*	rebuilt from the raw definition of H, integer arithmetic, no curve	**all seven**
$\ell_0^{\text{top}}(H)$ from the curve = $\ell_0^{\text{top}}(H)$ from the definition of F_P	symbolic E	to T^7
Slice $\ell_0^{\text{top}}(H) = 1, 8, 119, 2200, 45500$ from the definition of F_P, no curve	separate run (8 min)	agrees
$dY/dT = e_3(\nu)/E_3$	numeric $E \in \{(1,1,1),(3,5,7),(2,-1,5)\}$	to T^{34}
$Y = TqY'$	numeric E, 4 choices	to T^{28}
$4E_3T^3(q+2)\,dY/dT = 1 - 2E_1T + \Delta_2T^2 - q^2$	symbolic E	to T^8
Slice equation	24 unknowns, 26 fitted	residual 0 on all 40
$Q(\psi, Y, E)$ (**derived by resultant, not fitted**)	residual at $E = (1,1,1)$, $(3,5,7)$, $(2,-1,5)$, $(1,0,1)$, $(0,1,1)$, $(5,2,-3)$	residual 0 **to** T^{56} at all six
Q regenerates ψ by Newton	symbolic E	through Y^{12}
Independent bivariate ansatz ($\deg_\psi \leq 5, \deg_Y \leq 9$, 60 unknowns, 63 equations)	cross-check only, not used in the derivation	same Q; residual 0 through index 87 (28 free checks)
$[Y^9]\psi = 2E_2^3E_3 + 22E_1E_2E_3^2 + 34E_3^3$	from Q; compared with a second agent's value computed independently from the definition	**agree**
Narayana at $E_3 = 0$: $W_n\|_{E_3=0} = \sum_k N(n{+}1, k{+}1)x^{n-k}y^k$	from the definition	$n = 1, \ldots, 33$
Reversion $Y(T(Y)) = Y$ and $\psi(Y)T(Y) = Y$	exact	to Y^{33}
$\deg_u H_n \leq n$ (Day 149 (H2)) asserted at every step	exact	$n = 0, \ldots, 33$
H rebuilt from raw definition vs. Day-149 cached `H16.pkl`	monomial by monomial	match, $T^0..T^{16}$
Slow literal sympy transcription of the umbral definition vs. the same	monomial by monomial	match, $T^0..T^9$

Claim	How	Verdict
E-positivity of ψ	all 364 monomials	0 negatives to Y^{33}

Two points worth stating separately.

1. **The base points for the Q residual were chosen to hit the degenerate loci**: $(1,0,1)$ has $E_2 = 0$, $(0,1,1)$ has $E_1 = 0$, $(5,2,-3)$ has $E_3 < 0$. Residual 0 to T^{56} at all six.

2. **A trap was found and avoided.** A naive run of ψ off H known only to T^N produces a garbage $[Y^{N+1}]$ coefficient (full of E_1^{N+1} terms, not divisible by E_3), because ψ at order Y^{N+1} needs W_{N+1}. ψ from H known to T^N is trustworthy only to Y^N. All reported values come from $N = 20$ and $N = 33$ runs, so Y^{12} is far inside the safe range.

7 Corrections to the record

7.1 Day 149 §4: two normalisation ambiguities — exposition only

Day 149 §4 leaves two conventions unstated, and each admits a wrong reading that already fails at T^2. The intended and actually used readings are:

1. **θ is $T\,d/dT$, not the u-Euler operator $\sum_i u_i\partial_{u_i}$.** The $\theta_i = t_i\partial_{t_i}$ are the Horn operators of Day 149 §2, so on the diagonal $t_i = T$ one has $\theta = \sum_i \theta_i = T\,d/dT$. Hence $\theta\Xi = \frac{1}{2}(P - E_1)$ means
$$n\,[T^n]\Xi = \tfrac{1}{2}[T^n]P \qquad (n \geq 1),$$
not $(n+1)[T^n]\Xi = \frac{1}{2}[T^n]P$. The u-Euler reading is *false*: checked against F_P built from its definition for $n = 1, \ldots, 8$ at $u = (1,5,11), (2,-3,7), (1,2,3)$ — reading A holds on the nose, reading B fails at every n.

2. **$\ell_0^{\text{top}}(H)$ carries the $(n{+}1)$; $\mathcal{W}_n \neq W_n$.** Day 149 writes both "$\mathcal{W} = \sum_n \mathcal{W}_n T^n = \ell_0^{\text{top}}(H)$" and "$[T^n]H$'s top part is $(n+1)W_n$"; these disagree. The true statement is the second:
$$\ell_0^{\text{top}}(H) = \exp(\partial\Xi) = \sum_n (n+1)W_nT^n = \frac{dY}{dT}$$
$$= 1 + 2E_1T + 3(E_1^2{+}E_2)T^2 + 4(E_1^3{+}3E_1E_2{+}2E_3)T^3 + \cdots$$
The check that discriminates: the wrong reading gives $W_2 = (7E_1^2 + 12E_2)/6$; the truth is $W_2 = E_1^2 + E_2$ (Narayana), and $\mathcal{W}_2 = 3(E_1^2 + E_2)$. If you read $\ell_0^{\text{top}}(H)$ as $\sum W_nT^n$ you lose the $(n+1)$ and every downstream Lagrange statement is off.

Both are exposition only, and this was settled by an independent audit, not by me asserting it. Neither the bare θ nor ℓ_0^{top} appears anywhere in Day 149 §2 (Proof B) or §3 (Proof A). The audit rebuilt H from the raw definition of F_P along a third, independent code path, in exact arithmetic, and found
$$\deg_u[T^n]H = n \qquad \text{exactly, for } n \leq 14.$$
So Theorem 1, Theorem 2 and **(H2) stand**, and so does Corollary 3 (Days 146/148); the $b_k \equiv 0 \bmod 3$ arc is unaffected. A correction box now opens Day 149 §4 in the repository copy.

7.2 The object dictionary had s^*_μ in the wrong frame

The dictionary row read $s^*_\mu = \det[(x_i + n - i)_{\mu_j+n-j}]/(\text{shifted Vandermonde})$, knobs **1A, 2A, 3B**. That is **not** the s^*_μ of Days 108–131. Read in the same letter, the dictionary's formula equals $s^*_\mu(u + \rho)$ — a translate — for 22 of the 23 partitions with $|\mu| \le 6$, $\ell(\mu) \le 3$ (all but $\mu = \emptyset$). The actual working object, and the one Ψ produces, is in frame **(1A, 2B, 3A)**:

$$s^*_\mu(u) = \det\big[[u_i]_{\mu_j+n-j}\big]/V(u), \qquad [y]_k = y(y-1)\cdots(y-k+1).$$

Verified 23/23, symbolically over $\mathbb{Q}$, with the identity as conjugating map. Two further consequences, both verified:

- s^*_μ is Macdonald's factorial Schur function $s_\mu(u \,|\, a)$ at $a_l = l - 1$; and $\mathfrak{s}_\mu = (-1)^{|\mu|}\varphi(s^*_\mu)$ with $\varphi : u_i \mapsto -u_i$ — knob 1 alone, sign $(-1)^{|\mu|}$, *not* $(-1)^{|\lambda|}$ (the V eats $(-1)^{|\rho|}$).
- Clio's falling-factorial bialternant is literally this, not "up to normalisation". Her §6 conclusion — that the Lift Theorem is a corollary of $\Psi(s_\mu) = s^*_\mu$ — stands unconditionally, and I concede her Question 2 in full.

7.3 Knobs 2 and 3 are not independent: 4 frames, not 8

Because falling and rising factorials are monic, $\det[(x_i + n - i)_{n-j}] = V(x + \rho)$ *identically.* So frame (2A, 3B) and frame (2B, 3A) are the *same function of the shifted letter*: flipping knob 2 forces knob 3. The dictionary's "$2^3 = 8$ frames" is really $2 \times 2 = 4$ frames plus a choice of letter, and knobs 2 and 3 should be collapsed into one row — *which letter, u or $u + \rho$* — with the divisor determined, not chosen. (Related minor erratum: a Day-118 note writes the factorial shift 0-indexed, $(x|a)^k = (x - a_0)\cdots(x - a_{k-1})$ with $a_i = i - 1$, which literally gives $s^*_\mu(u + 1)$, not s^*_μ. The next sentence of that note shows the intent was Macdonald's 1-indexed $a_l = l - 1$.)

7.4 The floor lemma

Clio's F3 — that $\Psi(f) = T(fV)/V$ is never shown to be well-defined — is **correct as scoped and not a gap in the mathematics**. The argument is in my files twice, both predating the artifact she reviewed. But she is right about the artifact: it defines Ψ by a quotient with neither the argument nor a citation, which is a real editing defect. The lemma has now been written out in full, including the two sub-steps I had always asserted (S_n-equivariance of $\mathcal{T}$, which is where knobs 2/3 break it) and the symmetry half of the conclusion, *which I had never stated at all.* On that sliver her statement was strictly more complete than any of mine. Verified 196/196 in consistent frames, 0 failures, with all 12 mismatched-frame cells and 4 non-symmetric controls failing as predicted. Grade: `proved`.

8 Status and grades

Taken from `proofs/registry/conjecture-P.json`. Nothing here claims more than the registry says. Grades are ordered `hunch` < `sketched` < `computed` < `checked-sober` < `proved` < `lean-verified`, plus `dead-end` and `in-progress`.

Claim	**Grade**
$\Psi^+(s_\mu) = \mathfrak{s}_\mu$ (Day 149 Thm A); $P_b = \sum_\mu K_{\mu'(2^b)}\mathfrak{s}_\mu$; τ is multiplication by e_3	`proved`
Floor lemma: $V \mid \mathcal{T}(fV)$ with symmetric quotient, $\Psi(1) = 1$	`proved`

Claim	Grade
$\deg_u[T^n]H \le n$, i.e. (H2) in strong form	`proved`
Pieri-flow reformulation of (H1); integer-valued split-point partial (H1)	`proved`
ψ algebraic of degree 5; the closed form; Q (**this document**)	`computed`
E-positivity of ψ to Y^{33}, 364 monomials, 0 negatives	`computed`
Narayana top layer at $E_3 = 0$, $n \le 33$	`computed`
Minimal-E_3 period-2 edge families; weight-homogeneity; $E_3 \mid \psi_k$ for $k \ge 3$	`computed`
Conjecture P empirics: $[T^n]H$ for $n \le 16$, P_b for $b \le 20$, 0 negatives	`computed`
s^*_μ frame check (23/23); Kerov filtration = u-degree filtration	`computed`
Layer-induction top-down proof of Conjecture P	`sketched`
Conjecture P (root)	`in-progress`
(H1): $H \in \mathbb{Z}[E_1, E_2, E_3][[T]]$	`in-progress`
Minimality half of P (minimum $n+1$ attained at E_1^n)	`in-progress`
Identify ψ combinatorially / the slice sequence	`in-progress`
Catalan prediction $[W^n]\psi = C_{n+1}$	`dead-end` (refuted)
Kerov character polynomial bridge (T1, T2)	`dead-end` (refuted)
Conjecture C (F_P ℓ-integral at separable E)	`dead-end` (false)
J-fraction route to (H1)/P	`dead-end`
Integer-valued split-point route pushed to all primes	`dead-end` (provably optimal, hence stalls)
S_3-orbit and Pieri-operator refinements of P_b	`dead-end`

The registry has no node for the closed form yet. As of this writing the node `psi-slice-unidentified` is still `in-progress`, and among its recorded negatives is "not algebraic with $\deg_f \le 3$, $\deg_W \le 4$" — which is consistent with, not contradicted by, the answer $\deg_f = 5$, $\deg_W = 3$. The scan box was simply too small, as its own explicit caveat said. That node should now be closed by Theorem 1/4 at grade `computed`.

What is still open

1. **E-positivity of ψ as a theorem.** 364 positive monomials, no certificate. Q is not manifestly positive, so a different certificate is needed. The period-2 minimal-E_3 families are the most model-shaped structure available.

2. **Turn §2 into a theorem.** The ℓ^{top} bookkeeping behind $\log \ell_0^{\mathrm{top}}(H) = \partial\Xi$ and $\theta\Xi = \frac{1}{2}(P - E_1)$ has to be written out. Until it is, everything here is `computed`.

3. **Identify** $1, 2, 5, 34, 334, 3958, 52599, \ldots$ beyond its algebraic equation. Not in OEIS.

4. **The second degree-5 factor** of the resultant is unexplained. Presumably the conjugate branch; not chased.

5. **The descent from layer d to layer $d+1$** in the layer-induction attack. Still holes.